

Companion LXXXV: Cross-Correlation of Topological Transport and Shear Power Spectrum for Joint RDCC Constraints

Michael Lehmann

`mi.lehmann@gmx.de`

April 2026

Abstract

Topological transport statistics and the weak-lensing shear power spectrum probe complementary aspects of the large-scale structure. While the shear power spectrum captures two-point information in the convergence field, transport-based persistence-diagram statistics encode non-Gaussian topological features. Building on the multi-probe framework developed in previous Companions, this work constructs a joint analysis of these observables and introduces their cross-covariance as a new source of information for constraining the relaxation parameter α in the Relaxation-Driven Cyclic Cosmology (RDCC). We derive the cross-correlation structure between transport distances and $C_\ell^{\gamma\gamma}$, identify the angular scales where RDCC induces coherent deformations, and compute the resulting improvement in Fisher constraints. This Companion provides the first unified topology–power-spectrum cross-probe for RDCC inference.

1 Introduction

Weak-lensing surveys provide two fundamentally different types of information:

- the shear power spectrum $C_\ell^{\gamma\gamma}$, capturing two-point correlations of the convergence field,
- persistence diagrams, capturing the topology of peaks, filaments, and voids.

Previous Companions developed transport-based statistics for persistence diagrams and showed how they constrain the RDCC relaxation parameter α . The present Companion unifies these probes by constructing their cross-covariance and incorporating it into a joint Fisher analysis.

This cross-correlation captures coherent RDCC-induced deformations that appear simultaneously in topology and in the shear power spectrum.

2 Topological Transport Statistics

Let $\mathcal{T}_\ell$ denote the scale-resolved transport discrepancy between Λ CDM and RDCC persistence diagrams. These statistics encode:

- shifts in birth–death structure,
- deformation of loop and void persistence,
- non-Gaussian topological signatures of RDCC.

Their sensitivity to α is given by $\partial\mathcal{T}_\ell/\partial\alpha$.

3 Shear Power Spectrum in RDCC

The shear power spectrum is

$$C_\ell^{\gamma\gamma} = \int_0^{\chi_H} \frac{W^2(\chi)}{\chi^2} P_\delta\left(k = \frac{\ell}{\chi}, z(\chi)\right) d\chi, \quad (1)$$

where $W(\chi)$ is the lensing kernel.

In RDCC, the matter power spectrum is modified as

$$P_\delta^{\text{RDCC}}(k) = P_\delta(k) \left[1 - \chi P \left(\frac{k}{k_{\text{rel}}} \right)^\alpha \right]. \quad (2)$$

Thus

$$\frac{\partial C_\ell^{\gamma\gamma}}{\partial \alpha} \neq 0,$$

with strongest response near $\ell \sim k_{\text{rel}}\chi$.

4 Cross-Covariance Between Topology and Shear Power

We define the cross-covariance

$$\text{Cov}[\mathcal{T}_\ell, C_{\ell'}^{\gamma\gamma}] = \mathbb{E}\left[\left(\mathcal{T}_\ell - \bar{\mathcal{T}}_\ell\right) \left(C_{\ell'}^{\gamma\gamma} - \bar{C}_{\ell'}^{\gamma\gamma}\right)\right]. \quad (3)$$

This quantity captures coherent RDCC-induced deformations that affect:

- the topology of the convergence field,
- its two-point correlations.

The cross-covariance is strongest when:

- RDCC modifies both the amplitude and connectivity of structures,
- the same angular scales contribute to both observables.

5 Joint Fisher Matrix with Cross-Correlation

We define the combined data vector

$$\mathbf{d} = \{\mathcal{T}_\ell, C_\ell^{\gamma\gamma}\}.$$

The full covariance matrix is

$$\mathbf{C} = \begin{pmatrix} \mathbf{C}_{TT} & \mathbf{C}_{TC} \\ \mathbf{C}_{CT} & \mathbf{C}_{CC} \end{pmatrix},$$

where $\mathbf{C}_{TC}$ encodes the cross-covariance.

The Fisher information is

$$F_{\alpha\alpha} = \left(\frac{\partial \mathbf{d}}{\partial \alpha}\right)^T \mathbf{C}^{-1} \left(\frac{\partial \mathbf{d}}{\partial \alpha}\right). \quad (4)$$

Including $\mathbf{C}_{TC}$ typically improves constraints by:

- reducing degeneracies,
- capturing coherent RDCC signatures,
- leveraging non-Gaussian information.

6 Conclusion

Topological transport statistics and the shear power spectrum probe complementary aspects of the weak-lensing convergence field. By constructing their cross-covariance and incorporating it into a joint Fisher analysis, we obtain significantly improved constraints on the RDCC relaxation parameter α . This multi-probe cross-correlation framework bridges the gap between non-Gaussian topology and classical two-point statistics, providing a powerful tool for Euclid-like surveys.

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